

Problem 1.

$$\tau = I\alpha = I \frac{\Delta\omega}{\Delta t} = 168.75 \text{ N}\cdot\text{m}$$

Problem 2.

The correct answer is B.

Problem 3.

The correct answer is E.

Problem 4.

a) See solution.

$$\text{b) } a = \frac{2}{3} \left(\frac{F \sin \theta}{m} - g \right)$$

$$\text{c) } \mu_s \geq \frac{1}{3} \left(\tan \theta - \frac{mg}{F \cos \theta} \right)$$

Problem 5.

$$\text{a) } \omega_1 = \sqrt{\frac{3g}{L}}.$$

$$\text{b) } \omega_2 = \frac{2}{3} \sqrt{\frac{3g}{L}} = \frac{2}{\sqrt{3}} \sqrt{\frac{g}{L}}$$

Problem 6.

$$\text{a) } \omega_{b+} = \frac{3}{2L} \sqrt{2gh}$$

$$\text{b) } h = \frac{2}{3} L.$$

Problem 7.

a) The correct answer is A and D.

b) The correct answer is A and C.

$$\text{c) } a_x = -r\alpha$$

$$\text{d) } a_x = \frac{2}{3} g \sin \theta$$